Nirbhay Singh

Machine Learning | GenAI | LLMOps
MLOps | Vertex AI | SageMaker

CONTACT

- @ nirbhaysingh1@live.com
- in linkedin.com/in/nirbhaysingh1
- GH github.com/nirbhays
- M medium.com/@nirbhaysingh1
- W cloudtoai.in
- C credly.com/users/nirbhaysingh
- L Warsaw, Poland

CORE SKILLS

AI / ML / GENAI

Vertex AI, SageMaker, LLMs (GPT-4, Claude, Gemini), RAG, Prompt Engineering, Agentic AI, LangChain, Vector DBs

MLOPS & LLMOPS

Model Registry, Feature Store, ML Pipelines, A/B Testing, Model Monitoring, Data Drift Detection

DATA & ANALYTICS

BigQuery, Redshift, EMR, Spark, Enterprise Search, Data Pipelines, ETL

CLOUD

GCP, AWS, Azure, Multi-Cloud Architecture

INFRASTRUCTURE

Terraform, Kubernetes, Docker, CI/CD, GitOps

LANGUAGES

Python, Bash, SQL, YAML, HCL

EDUCATION

MTech (Master of Technology)

Birla Institute of Technology

LANGUAGES

English	Professional
Hindi	Native

Nirbhay Singh

Cloud & AI Architect | ML/AI & Generative AI Specialist

11+	14+	EUR 4M+	1,000+	60%
Years Experience	Certifications	Cloud Savings	Containers Orchestrated	AI Cost Reduction

PROFESSIONAL SUMMARY

Cloud & AI Architect with 11+ years architecting production-grade ML/AI systems across AWS, GCP, and Azure. Currently leading Generative AI and LLM integration at Bosch Europe -- designing enterprise search, document intelligence, and Agentic AI automation platforms adopted across multiple business units. Deep hands-on expertise in Vertex AI, SageMaker, RAG architectures, prompt engineering, and MLOps pipeline design. Built open-source AI cost intelligence tools (TokenMeter) achieving 60% LLM spend reduction. Google-certified Professional Machine Learning Engineer and Generative AI Leader with a strong track record of translating ML research into production-ready enterprise systems.

KEY ACHIEVEMENTS

- Architected LLM-powered enterprise search platform serving 15+ Bosch Europe business units using Vertex AI, RAG, and Agentic AI
- Designed end-to-end MLOps pipelines: training, evaluation, registry, deployment, monitoring with automated retraining and A/B testing
- Built TokenMeter (open-source) -- LLM cost intelligence tracking tokens across OpenAI, Anthropic, Google with smart routing saving 60%
- Deployed production ML serving on GKE, EKS, AKS with canary deployments, model versioning, and real-time inference at scale
- Created ShieldIaC -- AI-powered security scanner using GPT-4.1 for intelligent fix suggestions across 9 compliance frameworks

PROFESSIONAL EXPERIENCE

Cloud & AI Architect — Dec 2022 - Present

Robert Bosch (Bosch Polska) Warsaw, Poland

- Spearheaded GenAI adoption across Bosch Europe: integrated LLMs and Agentic AI into enterprise search, deploying on Vertex AI, BigQuery, and Cloud Run
- Designed RAG-based document intelligence platform processing 500K+ documents with semantic search across 15+ business units
- Established LLMOps framework: prompt versioning, evaluation pipelines, cost tracking, and guardrails for production deployments
- Led strategic AI partnership between Google Cloud, AWS, and Bosch Poland -- defined AI/ML architecture standards
- Implemented FinOps for AI/ML workloads, contributing to EUR 4M+ total cloud savings through GPU scheduling and preemptible instances

Senior DevOps Consultant (ML Platform) — Jun 2021 - Nov 2022

McAfee

- Deployed production ML serving on GKE and EKS with auto-scaling, canary deployments, and real-time inference
- Built ML monitoring: performance tracking, data drift detection, and automated alerting with Prometheus and Grafana
- Automated ML infrastructure with Terraform IaC, reducing deployment time from days to hours (50%+ improvement)

Senior DevOps Engineer / Data & AI Architect — Dec 2018 - Jun 2021

Deloitte

- Built data & AI platforms on SageMaker, EMR, and Redshift for model training, batch inference, and Fortune 500 analytics
- Orchestrated 1,000+ containers using AWS Batch, Spot Instances, and EKS for distributed ML training

- Reduced ML infrastructure costs by $45,000 through Spot Instance optimisation and GPU right-sizing

Senior Cloud Infrastructure Engineer

Nov 2015 - Dec 2018

PTC

- Designed fault-tolerant AWS architectures (EC2, RDS, S3, VPC) supporting data ingestion pipelines and early ML workloads
- Led cloud migration of 11,000+ server infrastructure, enabling data teams to adopt cloud-native ML and analytics tooling

System Engineer

Mar 2013 - Jul 2014

IBM

- Infrastructure operations, P1/P2 incident response, and root cause analysis for production enterprise systems

OPEN-SOURCE PROJECTS

TokenMeter -- Cost Intelligence Layer for LLM Apps

>> github.com/nirbhays/tokenmeter

One-line Python integration tracking cost, latency, and tokens across OpenAI, Anthropic, Google. Smart routing saves up to 60% on AI spend. <5ms overhead.

STACK: Python, OpenAI API, Anthropic API, Google AI, Smart Routing, PyPI

ShieldIaC -- AI-Powered IaC Security Scanner

>> github.com/nirbhays/shieldiac

100+ security rules, 9 compliance frameworks (CIS, SOC2, HIPAA, PCI-DSS, NIST). GPT-4.1 fix suggestions for Terraform and CloudFormation misconfigurations.

STACK: Python, Terraform, CloudFormation, GPT-4.1, GitHub Actions

InfraCents -- Terraform Cost Estimates on Every PR

>> github.com/nirbhays/infracents

Open-source GitHub App posting real-time AWS+GCP cost estimates directly on pull requests. Zero configuration -- parses Terraform changes and queries live pricing APIs.

STACK: Python, Next.js, Terraform, GitHub App, AWS/GCP Pricing APIs

CERTIFICATIONS & CREDENTIALS

- **Google** Professional Machine Learning Engineer (Valid Dec 2027)
- **Google** Professional Cloud Architect (Valid Jun 2026)
- **Google** Associate Cloud Engineer (Aug 2023)
- **AWS** Solutions Architect - Associate (Jan 2021)
- **HashiCorp** Terraform Associate
- **Google** Generative AI Leader (Valid Aug 2028)
- **Google** Professional Cloud DevOps Engineer (Valid Dec 2027)
- **AWS** Solutions Architect - Professional (Feb 2020)
- **CKAD** - Certified Kubernetes Application Developer (Valid Sep 2026)
- **MCSA** - Microsoft Certified Solutions Associate (Apr 2015)

TECHNICAL WRITING & THOUGHT LEADERSHIP

- "Our AI Bill Was $4,800 Last Month -- Nobody Knew Why" | "How I Built an AI-Powered IaC Security Scanner"
  Published on Medium -- deep dives into LLM cost intelligence (TokenMeter) and AI-powered security scanning (ShieldIaC).
  https://medium.com/@nirbhaysingh1